Vireen Chowdary Vesangi

Data Analyst

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Summary

Data analyst with hands-on experience across the full analysis lifecycle: gathering and cleaning multi-source data, writing SQL and Python for extraction and transformation, applying statistical modelling and machine learning, and building dashboards that turn raw data into decisions. Internship experience in financial/operations analytics plus a portfolio of deployed, end-to-end data products. MBA in Information Systems & Analytics and B.Tech in Artificial Intelligence Engineering — combining technical depth with the business framing to make analysis actionable for stakeholders.

Technical Skills

Languages & Query: SQL (joins, aggregations, subqueries, window functions, query optimization), Python (Pandas, NumPy, scikit-learn, Matplotlib, Seaborn), R (basics), JavaScript (ES6+).

BI & Visualization: Power BI (DAX, Power Query, Data Modelling, Row-Level Security), Tableau (LOD expressions, parameters, interactive dashboards), Advanced Excel (Pivot Tables, VLOOKUP, formulas, VBA), Streamlit, Gradio.

Statistics & Machine Learning: Hypothesis testing, Linear & Logistic Regression, ANOVA, MANOVA, Chi-Square, t-test, Mann-Whitney U, Spearman correlation, K-Means clustering, KNN, Random Forest, time-series forecasting (Prophet), empirical-Bayes shrinkage, Monte Carlo simulation, model calibration & validation; SPSS, PSPP.

Data Engineering: ETL pipeline design (pandas), JSON parsing, groupby aggregation, feature engineering, categorical bucketing, data cleaning & validation, multi-currency / multi-source reconciliation, fact & dimension modelling, KPI definition.

AI & NLP: BERT-based NLP, local LLM workflows (Ollama, Pydantic schemas), OpenCLIP (zero-shot vision), YOLOv8 fine-tuning, EasyOCR, keyword-frequency text analysis.

Tools & Platforms: Git/GitHub, FastAPI, REST APIs, pytest, cloud deployment (Vercel, Render), Google Colab, uv/pip environments.

Experience

Business Analyst Intern (Data & Reporting) — **May – Jul 2025**

Symbiosys Technologies, Visakhapatnam

- Analyzed 1,500+ multi-currency invoices for an international IT-services firm under the VP of Operations; owned the full workflow from raw exports to leadership-facing insights.
- Cleaned and joined client and invoice datasets using SQL and Advanced Excel — resolving date-format inconsistencies, misspelled fields, and multi-currency normalization; built a Python real-time currency-conversion utility (forex API with fallback handling) to standardize INR/JPY/SGD/AUD invoices for consistent reporting.
- Built three interactive Tableau dashboards: revenue & services performance, overdue payments, and monthly/quarterly trends vs target — surfacing ~Rs 2.7 Cr in total overdue exposure, ~Rs 1.2 Cr concentrated in a single overseas market, and one client accounting for ~Rs 72 L in receivables.
- Ran statistical analysis in SPSS: descriptive statistics (average invoice ~Rs 8.36 L, range Rs 6,500 to Rs 32 L+), Spearman rank correlation (ρ = -0.001, p = 0.989) showing invoice size did **not** predict payment delay, and K-Means clustering (3 segments) on invoice value × payment terms.
- Findings shifted management attention from invoice value to client-behaviour patterns; identified elevated delinquency on 45-day payment terms and recommended credit-policy adjustments, presented in management reviews.

Artificial Intelligence Intern — **Feb – May 2023**

Ozonetel Communications Pvt. Ltd., Hyderabad

- Worked on Agent Assist, a production NLP product for contact-centre operations; supported BERT-based model development, validation, and tuning on unstructured customer-interaction data.
- Built the customer-facing front-end interface connecting LLM-driven backends to a usable UI — translating model outputs into insights business users could act on.

Portfolio Projects

Captain's Call — Probabilistic IPL Match Simulator (Deployed) — *Python, pandas, FastAPI, React*

- Built and deployed an end-to-end, data-driven T20 cricket match simulator — data engineering, statistical modelling, backend API, frontend UI, and cloud deployment (Render + Vercel) all solo.
- Engineered a pandas ETL pipeline over raw nested Cricsheet JSON: 1,244 matches / 295,733 ball-by-ball deliveries across 18 IPL seasons (2008–2026), transformed via parsing, groupby aggregation, feature engineering, and categorical bucketing across 4 eras and 7 innings sub-phases.
- Developed a probabilistic per-ball outcome model (6 canonical outcomes conditioned on batter, bowler, era, phase) with empirical-Bayes shrinkage (k=20) and multi-level hierarchical back-off across a 26,250-cell parameter grid per discipline; sparsity audit confirmed ~75% of audited cells data-rich.
- Built a Monte Carlo win-probability engine (2,000 simulations per match); diagnosed and fixed a systematic 64/36 estimation bias, restoring ~50/50 symmetry for identical line-ups — verified with a regression-test suite.

- Calibrated the intent-control layer to MAE ∼1.04pp against real attacking-situation data; documented known model limitations as a first-class artifact.
- Click here to play with the Simulation
- Click here to view the GitHub repo

GeoSpot — Multi-Signal Country Prediction from Images *PyTorch, OpenCLIP, YOLOv8, EasyOCR*

- Rebuilt a 2022 university project into a modern CV system fusing three signals — OpenCLIP zero-shot scene classification, EasyOCR script detection across 8 non-Latin scripts, and a YOLOv8 flag detector fine-tuned on synthetic flag-in-scene composites.
- Designed a trust-weighted fusion layer with per-module weights set from measured precision on a labeled test set (not intuition); updating weights from empirical evidence improved top-1 accuracy from 66.7% to 73.9% (top-3: 87.0%).

Pilot-to-Scale Decision Support — Power BI + GenAI Copilot *Power BI, Python, Streamlit, Ollama*

- Designed a multi-table Power BI dashboard (5 fact/dimension tables, 4 global pilots) with DAX measures, weighted readiness scoring, go/hold/stop decision logic, and fleet-level rollups for autonomous-robotics scale-up decisions.
- Built a human-in-the-loop local-LLM analytics copilot (Python, Streamlit, Ollama, Pydantic-validated schemas) converting unstructured operational text into structured, validated analytical artifacts — privacy-first, fully local inference.

Academic Projects

Airbnb Pricing Analysis — Cross-City European Market *Excel, PSPP, Tableau, Power BI*

- Analyzed a multi-city European Airbnb dataset (Amsterdam, Paris, Rome, Vienna, and more) covering weekday/weekend prices, room types, host status, and proximity metrics.
- Built a Linear Regression pricing model (bedrooms, person capacity, attraction index, distance to centre); validated significant price differences across room types with ANOVA and tested room-type vs superhost independence with Chi-Square.
- Segmented listings into 5 K-Means clusters (Budget to Ultra-Luxury; dominant budget cluster of 3,948 listings) and presented findings via Tableau and Power BI dashboards.

AR/VR Customer Experience — Business Research Methods *SPSS, Python (Colab), NLP*

- Designed a mixed-methods study (cross-sectional survey + retailer interviews) grounded in the Technology Acceptance Model and Schwartz's Paradox of Choice; collected 134 consumer responses via stratified sampling.
- Tested 4 hypotheses: MANOVA (Wilks' Lambda $p = 0.001$) on AR/VR usage vs brand loyalty/trust/privacy; multiple regression (Brand Loyalty model $R^2 = 0.448$, Trust strongest predictor); Mann-Whitney U (chosen over t-test due to mixed Likert scales); and one-way MANOVA ($p = 0.035$) confirming higher customization raised satisfaction **and** anxiety/fatigue — the paradox-of-choice effect.
- Ran Python NLP keyword-frequency analysis on retailer interview transcripts, surfacing adoption barriers: cost, staff training, and perceived complexity.

Retail Sales & Returns Analytics — 6-Dashboard Tableau Workbook *Tableau*

- Built a six-dashboard interactive workbook on the Sample Superstore dataset: time-trend analysis (2014–2017), category/sub-category performance, US geographic distribution, customer segment & shipping insights, and a two-part returns & discount impact analysis identifying products at risk.
- Used a wide chart vocabulary — tree maps, bullet charts, box-and-whisker, heat maps, bubble scatters, stacked bars, geographic maps — with LOD expressions and parameters.

Community Health Analytics — 5-Dashboard Power BI Workbook *Power BI, DAX*

- Built dashboards covering population health, patient risk stratification (top 10% high-risk patients by composite risk score), chronic disease monitoring (HbA1c, adherence vs complication risk), social determinants of health, and public-health surveillance across 17+ Indian state clusters.

Cesim Retail Simulation — MBA Phygital Lab *Business strategy, pricing analytics*

- Competed across six rounds of a multi-region retail simulation with team decisions on product pricing, placement, channel mix, marketing spend, and demand forecasting; analyzed round-by-round performance to iteratively optimize revenue and profitability.

Education

MBA — Information Systems, Analytics & International Business **2024 – 2026**
GITAM University, Visakhapatnam *CGPA: 7.6/10*

B.Tech — Artificial Intelligence Engineering (Robotics Specialization) **2019 – 2023**
Amrita Vishwa Vidyapeetham, Amritapuri *CGPA: 7.5/10*

Certifications

- Microsoft PL-300 Power BI Data Analyst (in progress)
- Microsoft Power BI Data Analyst (Coursera)
- Microsoft Engage 2021
- Shore Fest Datathon 1st Place Finish